

reaction-box-appendix

10. The Reaction Box Is a Familiar Quantum Gate in Disguise

Once the box is framed as a control bit plus two reversible input/output qubits, it turns out to be something already studied: the priority bit selects between the two flavors of the **CNOT** gate. The Coylean map is, on this reading, a 2D lattice of CNOTs with a classical control pattern.

10.1 The gate

Working from the truth tables in §3, the box implements a unitary

$U_P : |v\rangle |h\rangle \rightarrow |d\rangle |r\rangle$ parameterized by the classical priority bit P :

Priority V (vertical wins): $d = v, \quad r = h \oplus v$
Priority H (horizontal wins): $d = v \oplus h, \quad r = h$

In the computational basis ($|00\rangle, |01\rangle, |10\rangle, |11\rangle$):

$$U_V = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \quad U_H = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

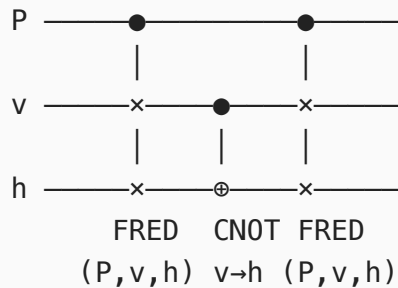
These are precisely the two CNOTs, distinguished only by which qubit plays the control role:

- $U_V = \text{CNOT}(v \rightarrow h)$ — v is control, h gets XORed.
- $U_H = \text{CNOT}(h \rightarrow v)$ — h is control, v gets XORed.

The "XOR toggle with priority breaking ties" rule, written out classically in §3, is literally the quantum CNOT. "Vertical wins" / "horizontal wins" just selects the control/target assignment.

10.2 The full controlled gate

Lifting the priority into a quantum register gives a 3-qubit gate U on $|P\rangle|v\rangle|h\rangle$ whose action is a direction-switching CNOT. A clean decomposition is a Fredkin–CNOT–Fredkin sandwich:



The outer Fredkins (controlled-SWAPs on v and h , conditioned on P) swap the two data qubits when $P = 1$. Sandwiched around a fixed-direction $\text{CNOT}(v \rightarrow h)$, the net effect is a CNOT whose direction is dictated by P : identity-on-direction when $P = 0$, flipped when $P = 1$. Verifying on the basis states reproduces the §3 tables.

10.3 Reading "4 reversible input/output pairs"

The phrase admits two readings, both pointing at the same physical fact.

Reading A — 4 ports. The box has four physical ports (N, W, S, E). Because the gate is involutive, each port carries one bit that can be read in either direction; the labels "input" and "output" are conventional, not physical. Four ports, four bidirectional signals.

Reading B — 4 basis-state pairings. The two-qubit gate is a permutation in S_4 . For each priority it is a product of disjoint 2-cycles (fixed points plus transpositions) — i.e. four reversible input/output state pairings.

Both readings reduce to the single fact $U_{P^2} = I$. The gate is unitary and Hermitian — a **quantum reflection**.

10.4 Why this matters

- A Hermitian unitary is the strongest form of self-inverse gate. The only single-qubit Hermitian unitaries are I , X , Y , Z , H and their signed variants. The CNOT is the canonical 2-qubit Hermitian unitary. The reaction box sitting exactly on this point of gate space is not coincidence — it is **what makes seam $\leftrightarrow$ patch invertibility hold**. The Coylean property in §4 is literally the Hermitian-unitary property in the 2-qubit setting.
- The Coylean map is a **classical-controlled CNOT lattice**: a 2D tiling of CNOTs with the direction at each cell selected by the 2-adic valuation of the absolute coordinates. The "priority field" of §5 is a classical control pattern over a CNOT lattice.
- This reframes the §7 measures in standard quantum terms. Bijectivity (§7.1) is unitarity. Involutivity (§7.2) is Hermiticity. The Hamming-distance and information-theoretic measures (§7.3, §7.4) are the natural relaxations: distance from a Hermitian unitary in operator norm, and the gate's classical channel capacity.
- The natural generalizations in §6 acquire concrete quantum names. Port arity $n = 3$ over $\{0,1\}$ with the strongest involutive candidate is the **Toffoli** gate. Higher-arity / multi-control Hermitian unitaries (controlled-Z, controlled-controlled-NOT, the full Clifford-group involutions) are the gate-level versions of the Coylean family. The search for "other algorithms with this symmetry" becomes the search for **classically-controlled lattices of Hermitian Clifford gates with a non-trivial priority assignment**.

Read this way, the Coylean map is the simplest non-trivial point in a large space already mapped out by reversible-computing and quantum-gate taxonomies — but combined with a number-theoretic priority assignment that those taxonomies do not, on their own, supply. The interesting object is the *combination*: the gate (Hermitian unitary) and the classical control field (2-adic valuation) together.